Geospatial Data Analysis for Smart Communities

Urban Infrastructure Stress Analysis: Mapping Traffic Population Dynamics in Zurich (2012-2025)

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GitHub: https://github.com/dydy2010/geospatial_analysis_hslu_group_project

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1 Introduction & Context

Zurich, Switzerland's largest city and a major European economic hub, is experiencing sustained **urbanization pressure** that increasingly tests the capacity and resilience of its transport infrastructure. Between **2012 and 2025**, the city's **population has continued to grow**, while the **road network has remained largely stable**. At the same time, **commuting patterns and mobility behaviors evolve**, concentrating demand on key **arterial roads** and **intersections**, especially during **peak hours**. This mismatch between growing demand and limited physical expansion contributes to **traffic congestion**, with downstream impacts on **travel time reliability**, **road safety**, **air quality**, and overall **network performance**.

For urban planning authorities, managing these challenges requires more than reacting to isolated bottlenecks. Decisions about **where to prioritize infrastructure investments**, **how to optimize traffic signal strategies**, and **which districts may require redesign or alternative mobility solutions** depend on a clear, data-driven understanding of how traffic behaves over time and across space. In particular, planners need to identify not only where volumes are high, but where **growth trends**, **directional imbalances**, and **recurring peak-demand patterns** signal structural stress on the network.

This project, *Urban Infrastructure Stress Analysis: Mapping Traffic Population Dynamics in Zurich (2012–2025)*, contributes to that goal by transforming raw traffic and population data into **actionable insights** for evidence-based urban planning. Using **geospatial data analytics**, the study links demographic dynamics with observed traffic volumes to highlight areas where infrastructure pressure is increasing most rapidly. A core contribution is the development of interpretable **key performance indicators (KPIs)**—including a district-level **Stress Index** that compares **population growth rates** to **traffic volume growth rates**—to support prioritization and long-term planning.

Beyond the stress index, the project examines **temporal traffic patterns** at multiple scales (**hourly, daily, and seasonal**) to detect peak demand periods and usage trends, and applies **clustering methods** to group counting sites by traffic profiles, revealing different "mobility zone" types across Zurich. Finally, results are communicated through **interactive geospatial visualizations and dashboards**, designed to help stakeholders explore patterns quickly and support planning discussions.

Overall, the work aims to bridge the gap between complex datasets and real planning needs: turning measurements into a coherent picture of **where**, **when**, and **how** Zurich's transport infrastructure is under growing stress—while also serving as a practical learning application of geospatial and analytical methods.

2 Data sources

We use three datasets, all retrieved from the Zurich Open Data Portal (Open Data Zürich).

Motorized individual traffic (MIV) counts — hourly values (2012–2025) Dataset provides hourly traffic measurements at multiple counting sites across the City of Zurich, available since 2012 (delivered daily, with hourly values per counting site).

Source: https://data.stadt-zuerich.ch/dataset/sid_dav_verkehrszaehlung_miv_od2031

Population statistics by quarter — monthly stock with demographic attributes Dataset provides population counts at the quarter (Stadtquartier) level, broken down by demographic dimensions (e.g., sex, age group, origin), enabling fine-grained spatial comparisons and trend analysis.

Source: https://data.stadt-zuerich.ch/dataset/bev_monat_bestand_quartier_geschl_ag_herkunft_od3250

Administrative boundaries (districts & quarters) — shapefiles / geodata Geospatial boundary dataset defining the official layout of Zurich's districts (Kreise) and statistical quarters, used for spatial joins, aggregation, and mapping.

Source: https://data.stadt-zuerich.ch/dataset/ktzh_stadtkreise_und_quartiere_zuerich_und_winterthur___ogd_

3 Tools & Technologies

We employed a Python-based geospatial analysis workflow for the project:

Data Processing: pandas for tabular data manipulation; GeoPandas for spatial operations and coordinate system transformations (Swiss LV95 to WGS84)

Analysis: scikit-learn for clustering algorithms; NumPy and SciPy for statistical computations

Visualization: Tableau

Geospatial Data: Zurich district and quarter boundary files (GeoJSON format) from opendata.swiss

4 Dataset & Preprocessing

Data Overview - Hourly **traffic count** data, **address** data (including city district and quarter) and **population** data from the City of Zurich's open data portal was collected. - The raw datasets consisted of **multiple CSV files** (**2012–2025**).

4.1 Raw Dataset: Traffic Data

Field Group	Columns
Identifiers & geometry	MSID, MSName, ZSID, ZSName, Achse, HNr, Hoehe, EKoord, NKoord, Richtung
Site metadata	Knummer, Kname, AnzDetektoren, D1ID–D4ID
Measurement facts	MessungDatZeit, LieferDat, AnzFahrzeuge, AnzFahrzeugeStatus

Because the raw dataset included a mixture of analytical attributes and highly technical metadata, each field was systematically examined and filtered using Python and the pandas library to assess its relevance for traffic-flow analysis.

4.1.1 Removed Fields

- **HNr**, due to inconsistent content.
- **D1ID–D4ID**, as these detector identifiers are technical metadata without analytical value.
- **LieferDat**, which is a delivery timestamp not required for traffic analysis.

All fields describing the **measurement location**, the **measurement configuration**, and the **traffic counts** themselves were retained. To improve clarity and support further processing and database integration, the remaining column names were **translated into English equivalents**. In a further preprocessing step, several **categorical values** contained in German were **translated to English**. After all preprocessing steps were completed, the resulting cleaned dataset contained **21,721,493 rows** and **14 columns**.

4.1.2 Cleaned Traffic Dataset Columns

- **measurement_site_id**: Unique technical identifier of the measurement site. A measurement site represents a specific traffic-flow direction or lane at a counting location.
- **measurement_site_name**: Technical name of the measurement site. In this dataset, this field contains "Unknown" for all entries.
- **counting_site_id**: Identifier of the counting site, representing the physical location where traffic measurements are collected.
- **counting_site_name**: Human-readable name of the counting site, describing the location.
- **axis**: Categorization of the counting site into a traffic axis (street).
- **position_description**: A textual descriptor indicating where along the street segment the measurement site is located. Contains "Unknown" for many entries.
- **east_coordinate**: The east coordinate of the measurement site in the Swiss CH1903+ / LV95 reference system.
- **north_coordinate**: The north coordinate of the measurement site in the Swiss CH1903+ / LV95 reference system.
- **direction**: The direction of traffic flow being measured (e.g., "inbound", "outbound").
- **signal_id**: Identifier of the associated traffic signal or intersection controller regulating traffic at the measurement site.
- **signal_name**: Name of the associated traffic signal or intersection.
- **num_detectors**: Number of detectors installed at the measurement site.
- **timestamp**: The timestamp indicating the start of the hourly measurement interval (ISO-8601 format).
- **vehicle_count**: The number of vehicles recorded during the hourly measurement interval.
- **vehicle_count_status**: Indicates how the vehicle count was produced: "Measured", "Missing", or "Imputed".

4.2 Raw Dataset: Quarters

The raw quarter dataset contained **address-level information** used by the City of Zurich for **administrative** and **statistical purposes**. The fields included:

Field Group	Columns
Identifiers	gwr_egid, objectid
Administrative units	stadtkreis, statistisches_quartier
Location metadata	adresse, hausnummer, lokalisationsname

Field Group	Columns
Area metrics	flaeche_real, flaeche_projektiert, flaeche_total
Spatial coverage counts	anzahl_fla_real, anzahl_fla_projektiert

4.2.1 Cleaning and Transformation

To link **traffic data** with **geographic units**, the raw fields required several preprocessing steps:

- **Address splitting and cleaning**: Street names and house numbers were embedded in a single free-text field (e.g., `"Heinrich-Federer-Strasse 12A"`, `"Widmerstrasse 88"`, `"Seeblickstrasse 17d"`). Using regular expressions, house numbers (including suffixes such as `12b`, `17d`) were removed, leaving a clean street-level identifier.
- **Data-type normalisation**: `statistical_quarter` was normalised as a trimmed string. Inconsistent labels such as `"Schwamend.-Mitte"` were harmonised to `"Schwamendingen-Mitte"` to ensure joinability with the population dataset.

4.2.2 Additional Standardisation for Street Name Matching

To ensure that traffic measurement locations could be linked reliably to the quarter dataset, several street names from the traffic dataset required manual standardisation. While most addresses were harmonised through regex-based cleaning, a small number of street names contained compound forms that could not be resolved automatically (e.g., combined street names, hyphenated patterns, or slash-separated names). These were corrected manually to create a consistent set of street identifiers.

Examples of manual mappings include:

- `Sood-/Leimbachstrasse` → `Soodstrasse`
- `Tobelhof-/Dreiwiesenstrasse` → `Tobelhofstrasse`
- `Manessestrasse - Schimmelstrasse` → `Manessestrasse`
- `Angererstrasse Tunnelstrasse` → `Angererstrasse`

Additionally, several counting site names referred to non-street features such as bridges, tunnels or motorway segments (e.g., `Quaibrücke, Milchbucktunnel, A1L`). For these cases, the closest corresponding street-level name from the quarter dataset was identified manually.

These manual adjustments were essential to achieve a **fully joinable** set of street names between the **traffic** and **quarter** datasets and ensured that **all counting sites** could be assigned to a **statistical quarter**.

4.2.3 Cleaned Quarter Dataset Columns

- **address**: Original full address as provided in the raw data.
- **house_number**: Extracted house number component (kept for completeness).
- **street_name**: Cleaned and standardised street name used as the spatial key.
- **city_district**: Official Zurich district number (1–12).
- **statistical_quarter**: Statistical quarter ("Statistisches Quartier") providing fine-grained spatial units.

The cleaned quarter dataset provides the **geographic reference layer** used to integrate **traffic measurements** with **demographic information**.

5 Data Processing and Methodology

This section describes the data processing pipeline and analytical methodology used to construct the Stress Index, which captures the relationship between population growth and traffic dynamics across Zurich's statistical quarters.

5.1 Traffic Data Processing

The raw traffic dataset provided by the City of Zurich consisted of approximately **3.7 GB** of data with over **20 million hourly observations** from permanent traffic counting stations.

5.1.1 Data Reduction and Cleaning

To ensure computational feasibility and analytical relevance, the following preprocessing steps were applied:

- **Column filtering**: Only essential variables were retained, including `measurement_site_id`, Swiss LV95 coordinates (east, north), `timestamp`, and `vehicle_count`.
- **Missing values**: Rows containing missing or invalid entries were removed.
- **Duplicate removal**: Duplicate observations were identified and dropped to avoid double counting.

5.1.2 Coordinate Reference System Transformation

Traffic measurement sites were originally recorded in the Swiss coordinate system **LV95 (EPSG:2056)**. For compatibility with spatial analysis and visualization tools, coordinates were transformed to **WGS84 (latitude/longitude)**.

5.2 Population Data Processing

The population dataset contained demographic breakdowns by age group, sex, and nationality at the quarter level.

5.2.1 Aggregation

To derive the total resident population per quarter and year:

- All demographic subgroups were **summed** to compute the **total population**
- Aggregation was performed for each **statistical quarter × year** combination

5.2.2 Annual Snapshot Selection

To ensure temporal consistency and avoid intra-year duplication:

- Only **January population snapshots** were retained

- This provides a stable annual population reference aligned with year-over-year growth calculations

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5.3 Geospatial Mapping of Traffic Sites to Quarters

To link traffic measurements to administrative units, a spatial join was performed using Zurich's statistical quarter boundaries.

- **Spatial predicate**: `within`
- **Input geometries**:
  - Point geometries for traffic measurement sites
  - Polygon geometries for statistical quarters
- **Outcome**:
  - **219 unique traffic measurement sites** were assigned to quarters
  - Each traffic observation inherits a corresponding `quarter` and `quarter_id`

This step enables aggregation of traffic counts at the quarter level.

5.4 Construction of the Stress Index

5.4.1 Temporal Aggregation

- **Traffic data**: Hourly vehicle counts were aggregated to **annual totals per quarter**
- **Population data**: Annual totals were already defined via January snapshots

The analysis focuses on the period **2012–2025**, corresponding to the availability of traffic sensor data.

5.4.2 Growth Rate Computation

For each quarter, year-over-year percentage growth rates were computed.

Population growth is defined as:

$$\text{Population Growth}_t = \frac{P_t - P_{t-1}}{P_{t-1}} \times 100$$

Traffic growth is defined as:

$$\text{Traffic Growth}_t = \frac{T_t - T_{t-1}}{T_{t-1}} \times 100$$

where P_t denotes the total population in year t, and T_t denotes the total traffic volume in year t.

Growth rates are undefined for the **first observed year** of each quarter due to the absence of a prior baseline.

5.4.3 Stress Index Definition

The **Stress Index** measures the imbalance between mobility demand and residential growth and is defined as:

$$\text{Stress Index} = \text{Traffic Growth (\%)} - \text{Population Growth (\%)}$$

- Positive values indicate traffic growth exceeding population growth
- Negative values indicate population growth exceeding traffic growth

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5.5 Classification of Quarters

Based on the Stress Index, quarters were categorized to support interpretation:

Category	Condition	Interpretation
Commuter Hub	Stress Index $> +5$	Strong external inflow of traffic
Balanced	$-5 \leq$ Stress Index $\leq +5$	Traffic and population grow in balance
Residential Zone	Stress Index < -5	Population growth dominates traffic

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5.6 Final Dataset

The final analytical dataset is exported as:

`stress_index_tableau_with_classes.csv`

It contains one observation per **quarter × year**, with the following key variables:

- `quarter`
- `quarter_id`
- `year`
- `total_population`
- `population_growth_pct`
- `total_traffic`
- `traffic_growth_pct`
- `stress_index`
- `category`

This dataset is optimized for downstream visualization and analysis in **Tableau**.

5.7 Methodological Considerations

- Extreme growth values (e.g. -100% or large positive rates) can occur when traffic volumes drop sharply or recover from low baselines
- Such values reflect real-world volatility (e.g. construction, sensor outages, exceptional events) and are retained as part of the analysis
- The Stress Index is intended as an **exploratory indicator** rather than a causal measure

6 Results

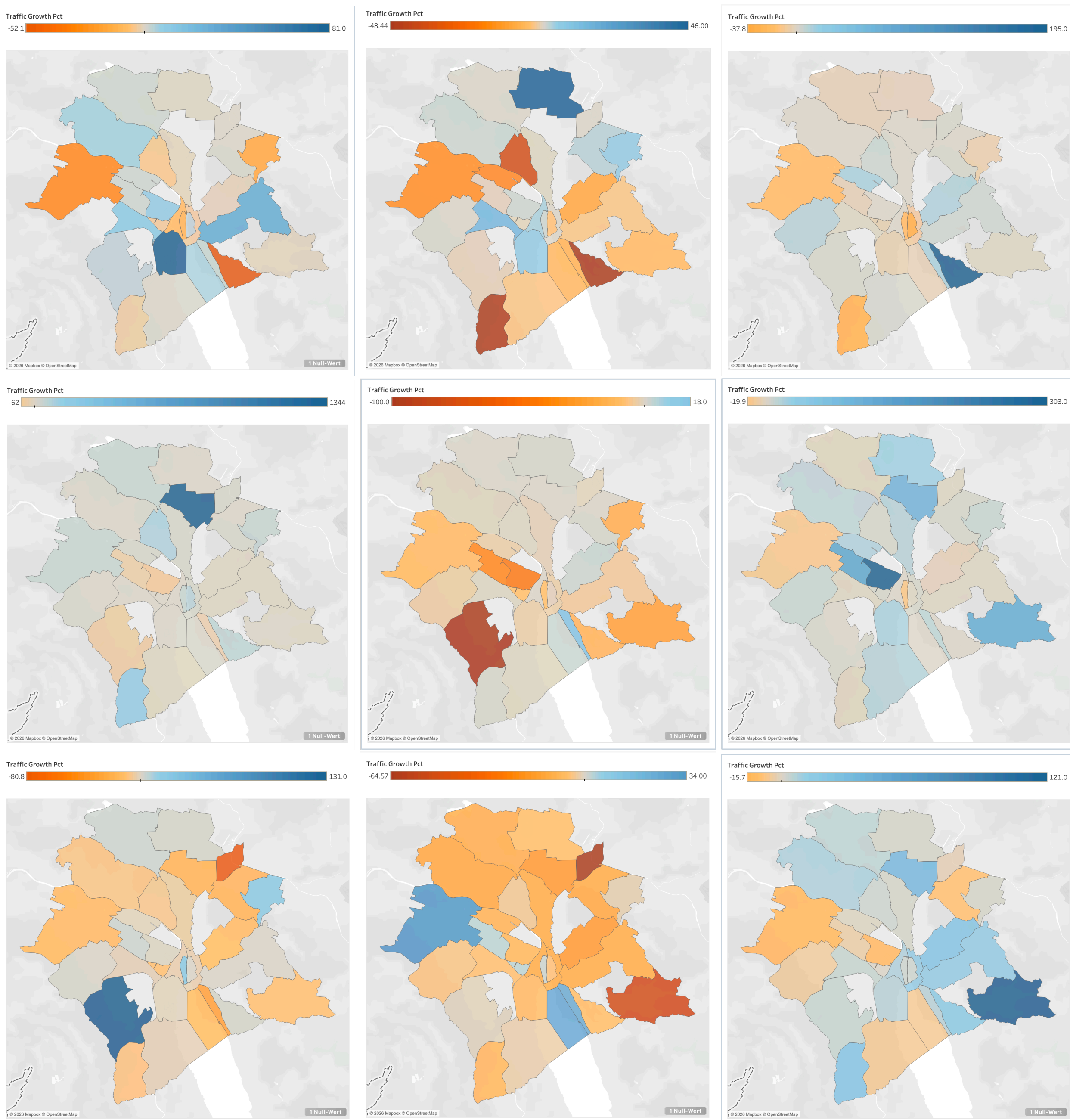
6.1 Traffic Growth

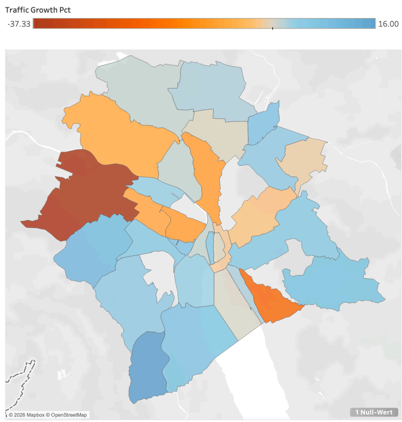
The traffic growth patterns across Zurich's 30 districts from 2012 to 2025 reveal a highly volatile and uneven landscape characterized by extreme fluctuations and significant geographic variation. Overall, traffic volumes experienced an average annual growth rate of 7.25%, though this figure masks substantial volatility (standard deviation of 79.86%) and a concerning reality: more than half of all year-over-year measurements (54.7%) actually showed traffic declines, with a median growth rate of -0.79%.

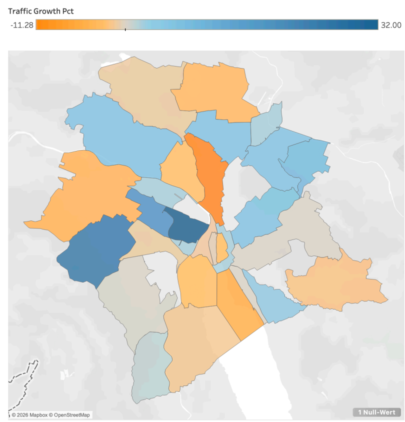
The temporal patterns show dramatic swings, including major spikes in 2016 (+53.90%), 2018 (+24.41%), and 2024 (+29.07%), followed by sharp corrections in 2017 (-8.70%), 2020 (-7.02% during COVID-19), and 2025 (-6.64%). Geographically, districts exhibit wildly different trajectories: peripheral and developing areas like Oerlikon (+139.87% average), Saatlen (+35.25%), and Albisrieden (+24.45%) experienced explosive traffic growth driven by infrastructure expansions and urban development, while central, established districts such as Werd (-2.59%), City (-2.26%), and Hochschulen (-1.90%) showed modest but consistent traffic declines, likely reflecting shifts toward public transportation, pedestrianization efforts, and changing urban mobility patterns.

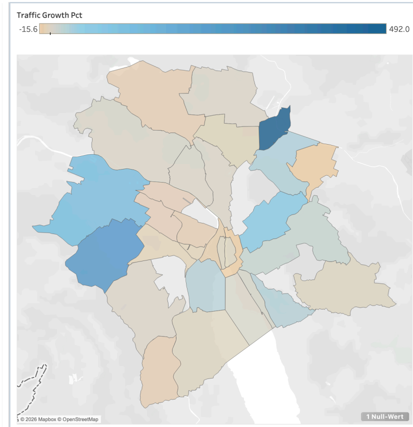
The extreme volatility in certain districts—particularly Oerlikon with individual years showing over 1300% growth—suggests either significant infrastructure changes, measurement methodology shifts, or data quality issues that warrant careful validation before drawing definitive policy conclusions.

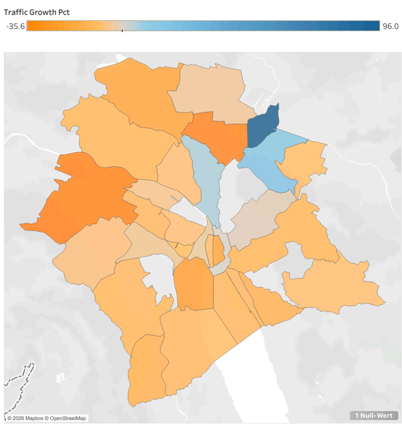
6.1.1 Traffic Growth 2013-2025, left to right, top to bottom











6.2 Population Growth

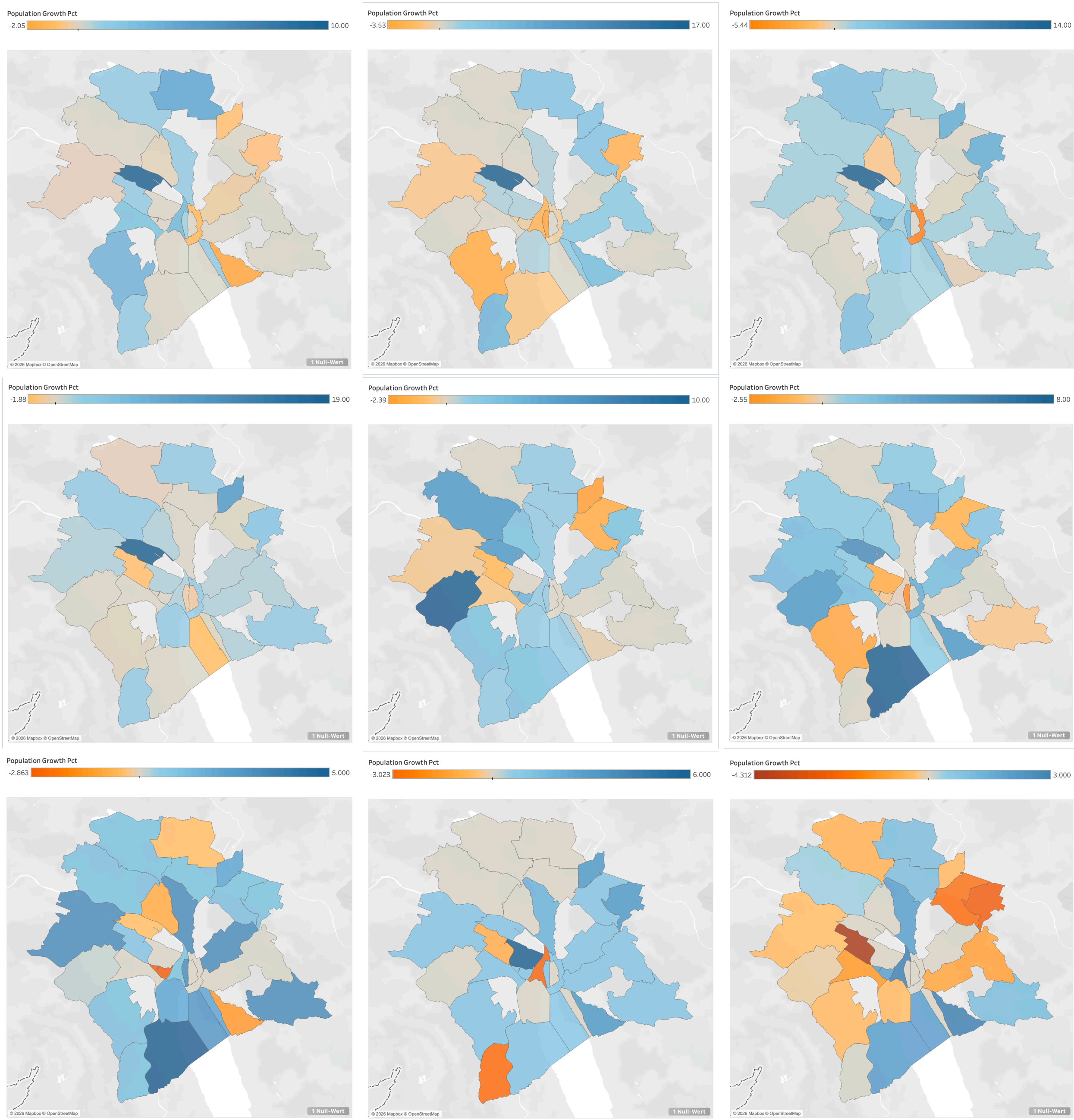
Population growth across Zurich's districts from 2012 to 2025 demonstrates a fundamentally different pattern than traffic growth—one characterized by steady, sustainable expansion with far lower volatility and more uniform geographic distribution. The city experienced consistent demographic growth with an average annual rate of 1.10% (median 0.88%), substantially more stable than traffic patterns as evidenced by the low standard deviation of 2.39%, and with the vast majority of district-year observations (71.7%) showing population increases.

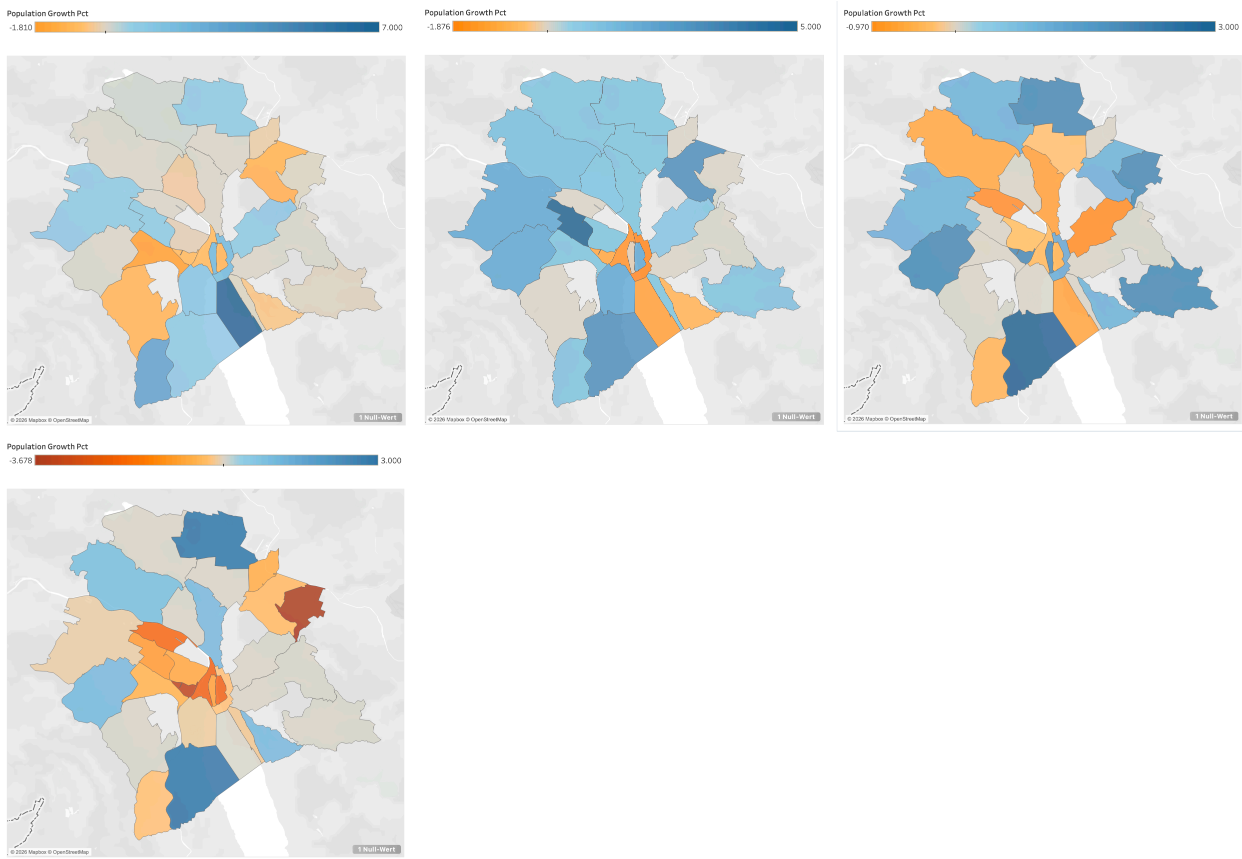
Over the 13-year period, Zurich's total population grew from 307,083 to 401,617 residents, representing a cumulative 30.8% increase, with particularly strong growth in the 2014-2015 period when the city added over 50,000 residents through urban development and migration. Annual growth rates remained relatively stable between 1.10% and 2.11% from 2013 to 2020, before moderating to 0.37%-1.59% in recent years and turning slightly negative in 2025 (-0.15%), possibly reflecting affordability pressures or pandemic-related demographic shifts.

Geographically, population growth concentrated in developing peripheral and waterfront districts like Escher Wyss (+5.43% average, experiencing dramatic transformation), Wollishofen (+2.45%), and Albisrieden (+2.08%), while established central districts such as City (+0.09%), Hochschulen (+0.15%), and Rathaus (+0.28%) showed minimal population change, suggesting these mature urban cores have reached build-out capacity while growth occurs in areas with available land for residential development.

Unlike the extreme district-level volatility in traffic patterns, population changes remained relatively predictable and policy-responsive, with the most volatile district (Escher Wyss) still showing a standard deviation of only 7.37%—far more manageable than the traffic volatility observed in comparable districts—indicating that urban planners have successfully managed demographic growth even as transportation infrastructure has struggled to keep pace.

6.2.1 Population Growth 2013-2025, left to right, top to bottom





6.3 Stress Index

The stress index, measuring the differential between traffic growth and population growth across Zurich's districts from 2012 to 2025, reveals a transportation system under significant and growing strain, with a concerning pattern of episodic crises punctuating an otherwise manageable baseline, though data limitations constrain the analysis with 7.2% of district-year observations lacking stress calculations due to missing baseline data (primarily the 2012 starting year where no prior year exists for comparison) and isolated instances of missing traffic data (notably Friesenberg 2017 with zero recorded traffic, and Oerlikon 2015 and Albisrieden 2014 with incomplete measurements).

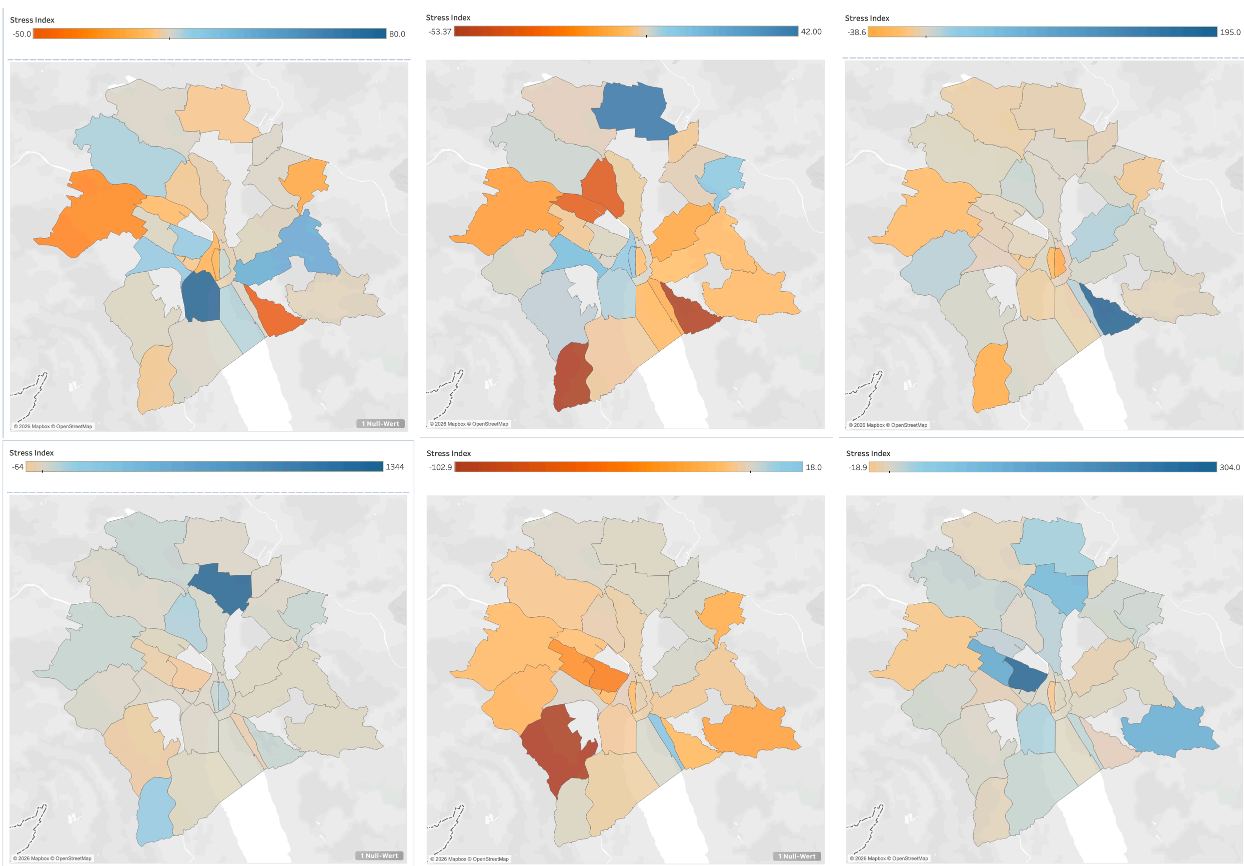
While the median stress index of -2.14% suggests that in typical district-years, population growth slightly outpaces traffic growth—indicating relative improvement in per-capita congestion—the mean of +6.15% and extreme standard deviation of 79.93% expose a reality dominated by severe outlier events that dramatically elevate average stress levels.

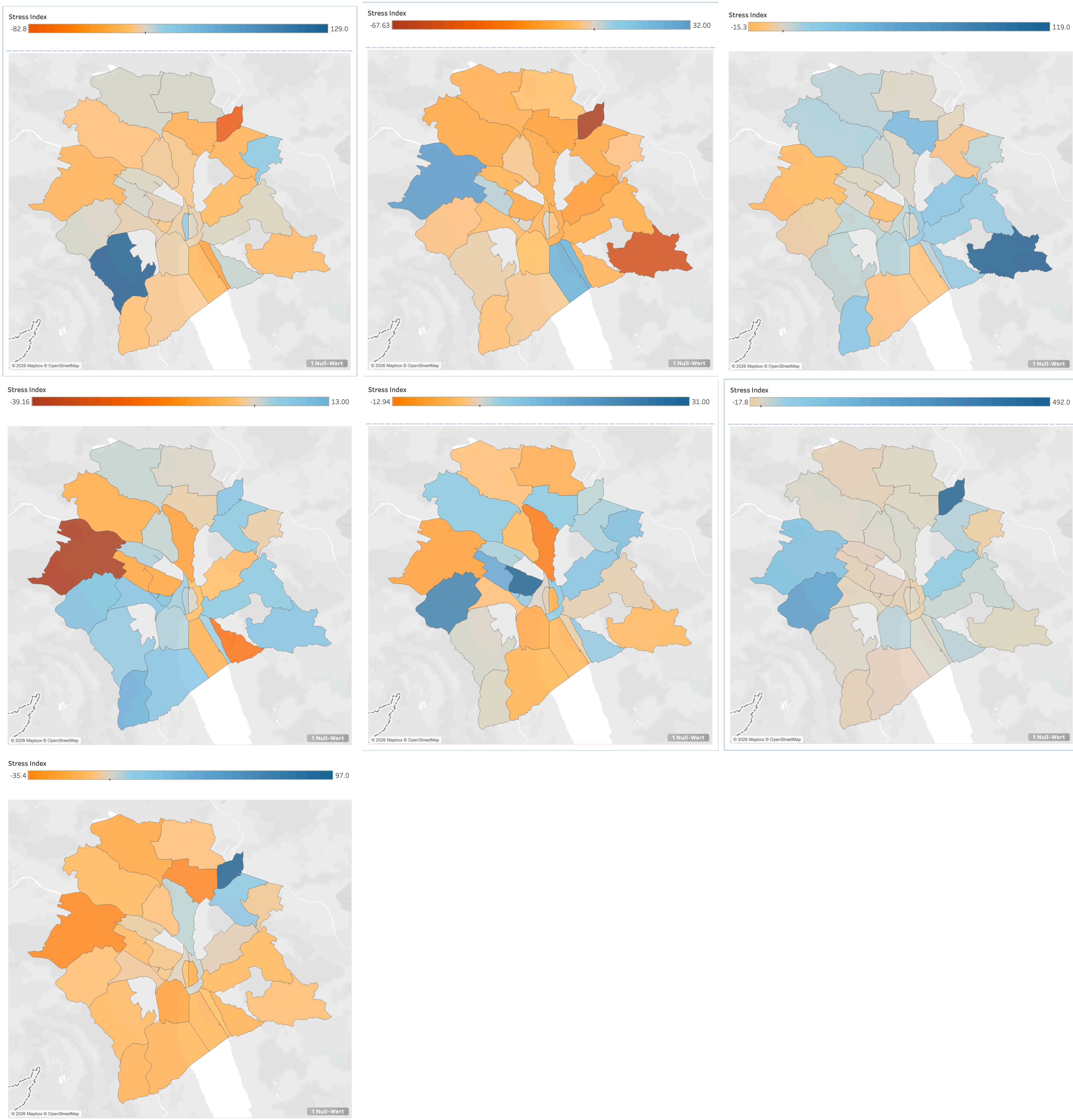
The temporal pattern shows three major crisis periods: 2016 (average stress +52.17%), 2018 (+22.99%), and most recently 2024 (+28.33%), driven by explosive traffic growth in specific districts that overwhelmed incremental population increases, while 2020 (-8.23%) and 2017 (-10.17%) provided temporary relief through COVID-19 disruption and traffic management interventions respectively.

Geographically, stress concentrates in a handful of transformation zones—Oerlikon (+138.95% average), Saatlen (+33.56%), Albisrieden (+22.37%), and Langstrasse (+13.99%)—where rapid infrastructure development, new road connections, or shifts in traffic routing created explosive growth that far exceeded population absorption capacity, with these districts experiencing multiple years of severe stress (>10%) despite otherwise stable demographic patterns. Conversely, central established districts like Escher Wyss (-5.79%), Wollishofen (-3.67%), and City (-2.35%) have achieved chronic improvement through sustained traffic reduction policies, pedestrianization, and public transport prioritization that outpaced their minimal population changes.

The distribution analysis reveals that 59.4% of district-years show negative stress (traffic growing slower than population), but the 40.6% showing positive stress includes a dangerous 13.6% with moderate-to-severe stress levels (>10%), and these crisis events disproportionately drive policy concerns, suggesting that Zurich's transportation challenge is not uniform congestion but rather managing extreme volatility and preventing catastrophic stress spikes in vulnerable districts undergoing rapid transformation.

6.3.1 Stress Index 2013-2025, left to right, top to bottom





7 Conclusion

7.1 Findings

This project analyzed the relationship between traffic growth and population growth across Zurich's quarters from 2012 to 2025. The main findings are:

Traffic is more volatile than population. While population grew steadily at around 1.1% per year, traffic volumes showed extreme swings. Some districts had years with over 100% traffic growth, followed by sharp declines. This makes traffic harder to predict and plan for.

Most quarters are actually improving. The median stress index of -2.14% means that in a typical year, population growth slightly outpaces traffic growth. About 59% of all district-year observations showed negative stress, meaning traffic is growing slower than population in most cases.

A few problem areas drive the overall concern. Districts like Oerlikon, Saatlen, Albisrieden, and Langstrasse showed very high stress values. These are transformation zones where new infrastructure or development created sudden traffic spikes. Oerlikon had an average stress of +139%, mainly due to one extreme year (2016) when traffic jumped over 1300%. This coincides with the completion of the Oerlikon Railway Station expansion as part of the Zürich Durchmesserlinie project. The station was expanded from six to eight tracks (Gleis 7 and 8 added) in late 2016. The Weinbergtunnel now connects Oerlikon directly to Zürich HB, and since December 2015, eleven S-Bahn lines serve the station. The expansion was so extensive that the historic MFO administration building had to be moved 60 meters westward in May 2012 to make room for the new tracks (Quelle: Wikipedia, https://de.wikipedia.org/wiki/Bahnhof_Zürich_Oerlikon).

Central districts are doing well. Established areas like City, Escher Wyss, and Wollishofen consistently showed negative stress. This suggests that pedestrianization, public transport, and traffic management policies are working in these zones.

COVID-19 provided temporary relief. The year 2020 showed a clear dip in traffic growth across all districts, resulting in lower stress levels. However, traffic rebounded in 2021-2024.

7.2 Limitations

This analysis has several limitations that should be considered when interpreting the results:

Missing quarters. Five out of 35 quarters (Oberstrass, Alt-Wiedikon, Hirslanden, Gewerbeschule, and one unknown) have no traffic sensor data. This means about 14% of Zurich cannot be analyzed with our method.

Data quality issues. Some traffic measurements show extreme values that are likely due to sensor problems rather than actual traffic changes. For example, Friesenberg 2017 recorded zero traffic, creating an undefined growth rate. The following year then shows "infinite" growth. These anomalies affect the averages.

Sensor installation effects. When new sensors are installed, traffic appears to increase dramatically overnight. The 1300% spike in Oerlikon (2016) is likely due to new sensors, not actual traffic changes. We cannot always distinguish real growth from measurement changes.

Traffic volume is not congestion. Our data counts the number of vehicles passing a sensor, not how long they are stuck in traffic. A road with high volume but good flow would look the same as a congested road in our data.

Population snapshots are from January only. We used January population figures for each year to avoid double-counting. This means seasonal population changes (students, tourists) are not captured.

Stress Index is exploratory. The classification thresholds (more than +5% for Commuter Hub, less than -5% for Residential Zone) are interpretive choices, not statistically derived cutoffs. Different thresholds would change the classification of some quarters.

Lack of local context. Our team is not from Zurich, so we lack the local knowledge that would help interpret some patterns. For example, understanding which streets had construction projects, which areas saw new developments, or which neighborhoods changed their traffic policies would add valuable context. A local urban planner would likely be able to explain many of the spikes and drops we observed.

Despite these limitations, the Stress Index provides a useful first look at which areas of Zurich face the biggest gap between population and traffic dynamics. Future work could incorporate congestion data, public transport usage, and more granular time-of-day analysis.

7.3 Learnings

This project deepened our skills in geospatial data analytics for urban planning by integrating Zurich's open datasets on traffic, population, and administrative boundaries into a single workflow. We learned to clean and aggregate large traffic time series, transform coordinates (LV95 $\rightarrow$ WGS84), and use spatial joins to link traffic counting sites to Zurich's statistical quarters. Finally, we strengthened our ability to create policy-relevant KPIs, especially the Stress Index (traffic growth vs. population growth), and to produce reproducible outputs ready for visualization and reporting.

8 Generative AI

Generative AI tools (including ChatGPT) supported our workflow by helping with coding, debugging, and exploring analysis approaches more efficiently. We used AI to speed up repetitive tasks (e.g., cleaning steps, feature ideas, and plotting logic), while keeping the final implementation under our control. To ensure reliability, we treated AI outputs as suggestions rather than ground truth: all generated code and conclusions were reviewed step-by-step, tested on our data, and validated to avoid errors propagating through the pipeline.